



Signals and Systems

Discrete Time Fourier Series

(lecture 27, Date:-10/04/2026)

By: Dr. Sajai Vir Singh

Example 16: Determine the inverse Fourier transform of following spectrum

$$X(\omega) = \left(\frac{1}{1 - \frac{1}{2}e^{-j\omega}} \right) \left(\frac{1}{1 - \frac{1}{3}e^{-j\omega}} \right)$$

$$\Rightarrow X(\omega) = \left(\frac{e^{j\omega}}{e^{j\omega} - \frac{1}{2}} \right) \left(\frac{e^{j\omega}}{e^{j\omega} - \frac{1}{3}} \right)$$

$$\Rightarrow \frac{X(\omega)}{e^{j\omega}} = \frac{e^{j\omega}}{(e^{j\omega} - \frac{1}{2})(e^{j\omega} - \frac{1}{3})}$$

Let $g = e^{j\omega}$

$$\begin{aligned}\Rightarrow \frac{X(\omega)}{g} &= \frac{g}{(g - \frac{1}{2})(g - \frac{1}{3})} = \frac{A}{(g - \frac{1}{2})} + \frac{B}{(g - \frac{1}{3})} \\ &= \frac{A(g - \frac{1}{3}) + B(g - \frac{1}{2})}{(g - \frac{1}{2})(g - \frac{1}{3})}\end{aligned}$$

$$\Rightarrow A(g - \frac{1}{3}) + B(g - \frac{1}{2}) \equiv g$$

 **To determine value of B, put $g = \frac{1}{3}$**

$$B(\frac{1}{3} - \frac{1}{2}) = \frac{1}{3} \quad \Rightarrow B = -2$$



To determine value of A, put $g=1/2$

$$A\left(\frac{1}{2} - \frac{1}{3}\right) = \frac{1}{2} \Rightarrow A = 3$$

So
$$\frac{X(\omega)}{g} = \frac{A}{\left(g - \frac{1}{2}\right)} + \frac{B}{\left(g - \frac{1}{3}\right)} = \frac{3}{\left(g - \frac{1}{2}\right)} + \frac{-2}{\left(g - \frac{1}{3}\right)}$$

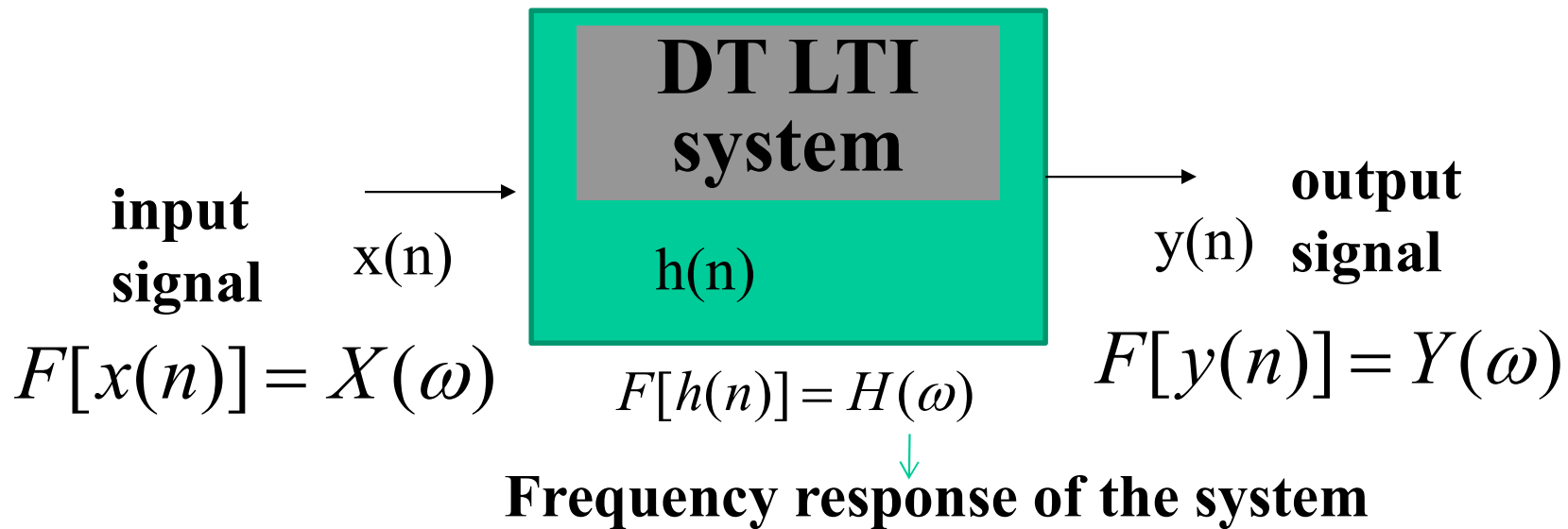
$$\Rightarrow \frac{X(\omega)}{e^{j\omega}} = \frac{3}{\left(e^{j\omega} - \frac{1}{2}\right)} + \frac{-2}{\left(e^{j\omega} - \frac{1}{3}\right)}$$

$$\Rightarrow X(\omega) = \frac{3e^{j\omega}}{\left(e^{j\omega} - \frac{1}{2}\right)} + \frac{-2e^{j\omega}}{\left(e^{j\omega} - \frac{1}{3}\right)} = \frac{3}{\left(1 - \frac{1}{2}e^{-j\omega}\right)} + \frac{-2}{\left(1 - \frac{1}{3}e^{-j\omega}\right)}$$

$$\Rightarrow X(\omega) = \frac{3}{(1 - \frac{1}{2}e^{-j\omega})} + \frac{-2}{(1 - \frac{1}{3}e^{-j\omega})}$$

So $x(n) = 3(\frac{1}{2})^n u(n) - 2(\frac{1}{3})^n u(n)$

System characteristics of Discrete Time LTI system (Frequency Response)



As we know

$$y(n) = x(n) * h(n)$$

$$F[y(n)] = F[x(n) * h(n)] \Rightarrow Y(\omega) = X(\omega)H(\omega)$$

$$\Rightarrow H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

$$Y(\omega) = X(\omega)H(\omega)$$

$$y(n) = F^{-1}[Y(\omega)] = F^{-1}[X(\omega)H(\omega)]$$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)}$$

$$h(n) = F^{-1}[H(\omega)] = F^{-1}\left[\frac{Y(\omega)}{X(\omega)}\right]$$

Example 17: Determine the frequency response and impulse of the LTI system described by following equation. Also find the response of the system when input applied is $x(n) = (1/3)^n u(n)$

$$y(n) - y(n-1) + \frac{3}{16} y(n-2) = x(n) - \frac{1}{2} x(n-1)$$

solution: $y(n) - y(n-1) + \frac{3}{16}y(n-2) = x(n) - \frac{1}{2}x(n-1)$

Taking Fourier Transform on both side

$$Y(\omega) - e^{-j\omega}Y(\omega) + \frac{3}{16}e^{-2j\omega}Y(\omega) = X(\omega) - \frac{1}{2}e^{-j\omega}X(\omega)$$

$$\Rightarrow Y(\omega)(1 - e^{-j\omega} + \frac{3}{16}e^{-2j\omega}) = X(\omega)(1 - \frac{1}{2}e^{-j\omega})$$

$$\Rightarrow H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{(1 - \frac{1}{2}e^{-j\omega})}{(1 - e^{-j\omega} + \frac{3}{16}e^{-2j\omega})}$$

Impulse response $h(n) = F^{-1}[H(\omega)]$

$$\Rightarrow H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{(1 - \frac{1}{2}e^{-j\omega})}{(1 - e^{-j\omega} + \frac{3}{16}e^{-2j\omega})} = \frac{(1 - \frac{1}{2}e^{-j\omega})}{(1 - \frac{1}{4}e^{-j\omega})(1 - \frac{3}{4}e^{-j\omega})}$$

Partial fraction

$$\frac{H(\omega)}{e^{j\omega}} = \frac{(e^{j\omega} - \frac{1}{2})}{(e^{j\omega} - \frac{1}{4})(e^{j\omega} - \frac{3}{4})}$$

Let $g = e^{j\omega}$

$$\frac{H(\omega)}{g} = \frac{(g - \frac{1}{2})}{(g - \frac{1}{4})(g - \frac{3}{4})} = \frac{A}{(g - \frac{1}{4})} + \frac{B}{(g - \frac{3}{4})}$$

$$A(g - \frac{3}{4}) + B(g - \frac{1}{4}) \equiv (g - \frac{1}{2})$$

On solving above for A and B, we get

$$A = \frac{1}{2} \quad B = 1$$

$$H(\omega) = \frac{A}{(1 - \frac{1}{4}e^{-j\omega})} + \frac{B}{(1 - \frac{3}{4}e^{-j\omega})} = \frac{1/2}{(1 - \frac{1}{4}e^{-j\omega})} + \frac{1}{(1 - \frac{3}{4}e^{-j\omega})}$$

Hence

$$h(n) = F^{-1}[H(\omega)] = \frac{1}{2}(\frac{1}{4})^n u(n) + (\frac{3}{4})^n u(n)$$

Out put response $y(n)$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{(1 - \frac{1}{2}e^{-j\omega})}{(1 - \frac{1}{4}e^{-j\omega})(1 - \frac{3}{4}e^{-j\omega})}$$

$$x(n) = (\frac{1}{3})^n u(n) \Rightarrow X(\omega) = \frac{1}{(1 - \frac{1}{3}e^{-j\omega})}$$

$$Y(\omega) = X(\omega)H(\omega) = \frac{1}{(1 - \frac{1}{3}e^{-j\omega})} \frac{(1 - \frac{1}{2}e^{-j\omega})}{(1 - \frac{1}{4}e^{-j\omega})(1 - \frac{3}{4}e^{-j\omega})}$$

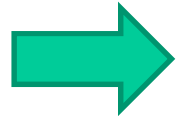
$$\frac{Y(\omega)}{e^{j\omega}} = \frac{(e^{j\omega} - \frac{1}{2})e^{j\omega}}{(e^{j\omega} - \frac{1}{3})(e^{j\omega} - \frac{1}{4})(e^{j\omega} - \frac{3}{4})}$$

Let $g = e^{j\omega}$

$$\begin{aligned} \Rightarrow \frac{Y(\omega)}{g} &= \frac{(g - \frac{1}{2})g}{(g - \frac{1}{3})(g - \frac{1}{4})(g - \frac{3}{4})} = \frac{A}{(g - \frac{1}{3})} + \frac{B}{(g - \frac{1}{4})} + \frac{C}{(g - \frac{3}{4})} \\ &= \frac{A(g - \frac{1}{4})(g - \frac{3}{4}) + B(g - \frac{1}{3})(g - \frac{3}{4}) + C(g - \frac{1}{3})(g - \frac{1}{4})}{(g - \frac{1}{3})(g - \frac{1}{4})(g - \frac{3}{4})} \end{aligned}$$

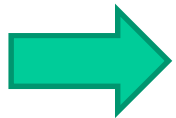
$$\Rightarrow A\left(g - \frac{1}{4}\right)\left(g - \frac{3}{4}\right) + B\left(g - \frac{1}{3}\right)\left(g - \frac{3}{4}\right) + C\left(g - \frac{1}{3}\right)\left(g - \frac{1}{4}\right) \equiv \left(g - \frac{1}{2}\right)g$$

Now find A, B, C



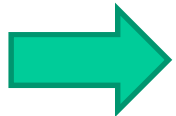
To determine value of B, put $g=1/4$

$$B = -3 / 2$$



To determine value of C, put $g=3/4$

$$C = 9 / 10$$



To determine value of A, put $g=1/3$

$$A = 8 / 5$$

Hence

$$Y(\omega) = \frac{A}{\left(1 - \frac{1}{3}e^{-j\omega}\right)} + \frac{B}{\left(1 - \frac{1}{4}e^{-j\omega}\right)} + \frac{C}{\left(1 - \frac{3}{4}e^{-j\omega}\right)}$$

$$\Rightarrow Y(\omega) = \frac{8/5}{(1 - \frac{1}{3}e^{-j\omega})} + \frac{-3/2}{(1 - \frac{1}{4}e^{-j\omega})} + \frac{9/10}{(1 - \frac{3}{4}e^{-j\omega})}$$

Hence

$$y(n) = F^{-1}[Y(\omega)] = \frac{8}{5} \left(\frac{1}{3}\right)^n u(n) - \frac{3}{2} \left(\frac{1}{4}\right)^n u(n) + \frac{9}{10} \left(\frac{3}{4}\right)^n u(n)$$

Example 18: Determine the frequency response and impulse of the LTI system described by following equation. Also find the response of the system when input applied is $x(n) = (1/2)^n u(n)$

$$y(n) - \frac{1}{4}y(n-1) - \frac{3}{8}y(n-2) = x(n) + x(n-1)$$

Ans

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{(1 + e^{-j\omega})}{(1 - \frac{1}{4}e^{-j\omega} - \frac{3}{8}e^{-2j\omega})}$$

Response of System when input applied is Eigen signals

$$y(n) = H(\omega_0)x(n)$$

Eigen functions are complex exponential and sinusoidal functions

$$x(n) = Ae^{\pm j\omega_0 n}$$



$$y(n) = AH(\omega_0)e^{\pm j\omega_0 n}$$

$$x(n) = A \cos \omega_0 n$$



$$\begin{aligned} y(n) &= AH(\omega_0) \cos \omega_0 n \\ &= A |H(\omega_0)| \cos(\omega_0 n + \angle H(\omega_0)) \end{aligned}$$

$$x(n) = A \sin(\omega_0 n + \theta)$$



$$\begin{aligned} y(n) &= AH(\omega_0) \sin(\omega_0 n + \theta) \\ &= A |H(\omega_0)| \sin(\omega_0 n + \theta + \angle H(\omega_0)) \end{aligned}$$

Example 19: Determine the Output response of the system with impulse response $h(n) = (1/2)^n u(n)$

When input is the complex exponential sequence given by

$$x(n) = Ae^{j\pi n/2}$$

Solu: $h(n) = (1/2)^n u(n)$

$$y(n) = AH(\omega_0)e^{j\pi n/2} = AH(\pi/2)e^{j\pi n/2}$$

$$\Rightarrow y(n) = A\left(\frac{2}{\sqrt{5}}e^{-j26.6^\circ}\right)e^{j\pi n/2}$$

$$= A\frac{2}{\sqrt{5}}e^{j(\pi n/2 - 26.6^\circ)}$$

$$H(\omega) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$


$$H(\pi/2) = \frac{1}{1 - \frac{1}{2}e^{-j\pi/2}} =$$

$$\frac{1}{1 + j1/2} = \frac{2}{\sqrt{5}}e^{-j26.6^\circ}$$

Example 20: Determine the Output response of the system in previous example to the input

$$x(n) = 10 - 5\sin\left(\frac{\pi}{2}n\right) + 20\cos(\pi n) + 3e^{j\pi n/2}$$

Solu:

$$y(n) = 10H(0) - 5H(\pi/2)\sin\left(\frac{\pi}{2}n\right) + 20H(\pi)\cos(\pi n) + 3H(\pi/2)e^{j\pi n/2}$$

$$\Rightarrow y(n) = 10H(0) - 5|H(\pi/2)|\sin\left(\frac{\pi}{2}n + \angle H(\pi/2)\right)$$

$$+ 20|H(\pi)|\cos(\pi n + \angle H(\pi)) + 3|H(\pi/2)|e^{j(\pi n/2 + \angle H(\pi/2))}$$

$$H(\omega) = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}$$

$$H(0) = 2, \dots, H(\pi) = \frac{2}{3}$$

$$H(\pi/2) = (2/\sqrt{5})e^{-j26.6^\circ}$$

$$\Rightarrow y(n) = 20 - \frac{10}{\sqrt{5}}\sin\left(\frac{\pi}{2}n - 26.6^\circ\right) + \frac{40}{3}\cos(\pi n) + \frac{6}{\sqrt{5}}e^{j(\pi n/2 - 26.6^\circ)}$$